

ModelArts

Service Overview

Issue	01
Date	2026-09-08



Copyright © Huawei Cloud Computing Technologies Co., Ltd. 2026. All rights reserved.

No part of this document may be reproduced or transmitted in any form or by any means without prior written consent of Huawei Cloud Computing Technologies Co., Ltd.

Trademarks and Permissions



HUAWEI and other Huawei trademarks are the property of Huawei Technologies Co., Ltd.

All other trademarks and trade names mentioned in this document are the property of their respective holders.

Notice

The purchased products, services and features are stipulated by the contract made between Huawei Cloud and the customer. All or part of the products, services and features described in this document may not be within the purchase scope or the usage scope. Unless otherwise specified in the contract, all statements, information, and recommendations in this document are provided "AS IS" without warranties, guarantees or representations of any kind, either express or implied.

The information in this document is subject to change without notice. Every effort has been made in the preparation of this document to ensure accuracy of the contents, but all statements, information, and recommendations in this document do not constitute a warranty of any kind, express or implied.

Huawei Cloud Computing Technologies Co., Ltd.

Address: Huawei Cloud Data Center Jiaoxinggong Road
Qianzhong Avenue
Gui'an New District
Gui Zhou 550029
People's Republic of China

Website: <https://www.huaweicloud.com/intl/en-us/>

Contents

1 What Is ModelArts?	1
2 ModelArts Functions	4
3 AI Development Basics	6
3.1 Introduction to the AI Development Lifecycle	6
3.2 Common Concepts of ModelArts	7
4 Security	9
4.1 Shared Responsibilities	9
4.2 Asset Identification and Management	10
4.3 Identity Authentication and Access Control	10
4.4 Data Protection	12
4.5 Auditing and Logs	12
4.6 Service Resilience	13
4.7 Security Risk Monitoring	15
4.8 Fault Recovery	15
4.9 Update Management	16
4.10 Certificates	17
4.11 Secure Boundaries	18
5 Notes and Constraints	21
6 Permissions Management	26
7 Billing Description	32
8 Quotas	33
9 ModelArts and Other Services	36

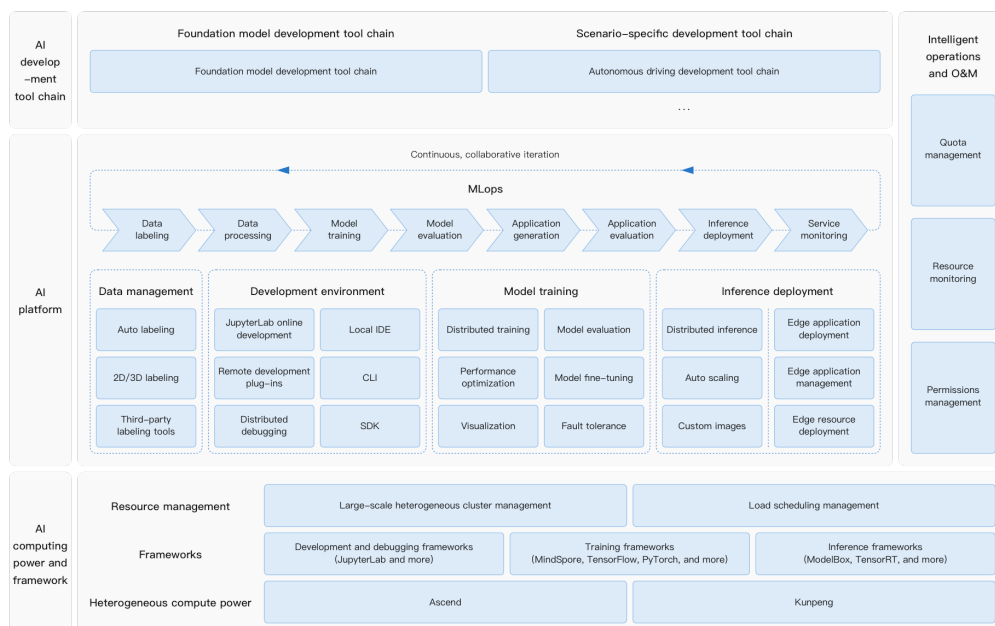
1 What Is ModelArts?

ModelArts is a one-stop AI development platform provided by Huawei Cloud. Dedicated to transforming AI technology into highly efficient and user-friendly productivity tools, ModelArts provides a comprehensive full-lifecycle toolchain. This includes algorithm development, model training, deployment, and resource management. The platform is designed to significantly lower the technical barriers and time costs of AI application development, allowing developers to focus entirely on business innovation. Furthermore, the underlying architecture of ModelArts is compatible with various heterogeneous compute resources and offers diverse resource scheduling capabilities. It provides comprehensive support for mainstream frameworks such as MindSpore, TensorFlow, PyTorch, Ascend-vLLM, LlamaFactory, and Verl, enabling developers to flexibly use powerful compute without needing to manage the underlying infrastructure.

Architecture

For details about the ModelArts architecture, see [Figure 1-1](#).

Figure 1-1 ModelArts architecture



- The compute layer offers a full range of Ascend hardware, ultra-large-scale cluster management, and resource scheduling and management. It supports popular AI development, debugging, training, and inference frameworks.
- The AI platform layer offers an end-to-end AI development toolchain, enabling you to develop and deploy models quickly and easily. It also enables efficient resource management and automatic fault recovery, speeding up AI model development, training, and deployment.
- The AI development toolchain layer offers an end-to-end model development toolchain, including high-quality open-source models and development suites that streamline the development process and reduce time to market.

Advantages

ModelArts has the following advantages:

Simplified and highly reliable model training

- **Zero-code/Low-code development:** Supports remote development via local IDEs with ModelArts plugins. Features built-in images for mainstream machine learning frameworks and an integrated architecture for seamless online/offline collaborative development and training. Supports distributed deployment and inference for large models.
- **High-efficiency optimization:** Capable of running ultra-large-scale distributed tasks. Provides O&M capabilities such as performance profiling and fault diagnosis within a stable, flexible, and high-performance training environment.

High-performance deployment and O&M

- Utilizes prefill-decode auto-scaling and ultra-dense deployment to maximize resource utilization.
- Provides multi-level rate limiting and token-level fast recovery to ensure the high availability of inference services.
- Supports multiple deployment modes, including real-time, batch, and edge inference.

Cost-effective compute

Offers various high-performance, cost-effective compute with flexible billing modes, including pay-per-use and yearly/monthly subscriptions. Includes comprehensive features for resource scheduling, task management, and real-time usage monitoring to optimize overall efficiency.

Efficient development and debugging tools

Supports remote development via local IDEs with ModelArts plugins. Features built-in images for mainstream machine learning frameworks and an integrated architecture for seamless online/offline collaborative development and training. Supports distributed deployment and inference for large models.

Access Methods

ModelArts provides multiple access methods tailored to different product forms.

- Console
ModelArts can be accessed via the management console, which provides a full suite of features including data connections, development environment, model training, model deployment, and model evaluation. You can complete the entire end-to-end AI development lifecycle directly within the console.
- APIs
If you want to integrate ModelArts into a third-party system for secondary development, use APIs to access ModelArts. For details about the APIs and operations, see [API Reference](#).
- Cloud native
If you are using a Lite Server, you can use an EIP to directly access the cloud server. For details, see [Using Lite Server](#).
If you are using a Lite Cluster, you can use native Kubernetes APIs to perform operations on the cluster. [Using Lite Cluster](#).

2 ModelArts Functions

ModelArts turbocharges compute with framework-level performance optimization, enabling faster training and more flexible deployment. It offers a comprehensive suite of features, including data preparation, model development and training, model deployment, live comparison, model debugging, resource management, and O&M and operations.

Table 2-1 ModelArts functions

Function Module	Description
Model development	ModelArts offers a flexible and open development environment that is ready to use without local configuration. It supports mainstream engines such as PyTorch, TensorFlow, and MindSpore. You can operate online via JupyterLab or perform remote development by connecting local IDEs (such as VS Code and PyCharm) through SSH.
Model training	As an enterprise-grade AI distributed training platform, ModelArts is fully compatible with industry-leading training frameworks and models. It supports one-click initiation of ultra-large-scale distributed tasks and provides a one-stop training solution featuring full-stack acceleration, long-term stability, and agile usability. It achieves a massive leap forward in both model performance and business iteration speed.
Model deployment	ModelArts is a high-reliability, next-generation distributed elastic inference platform that integrates distributed deployment, elastic inference, and intelligent scheduling. Deeply integrated with Huawei-developed and optimized vLLM inference framework, it supports one-click deployment of mainstream open-source models. It provides a more efficient, flexible, and easy-to-use large model inference solution, helping enterprises quickly implement AI applications.

Function Module	Description
Resource management	Provides various compute resources and associated scheduling capabilities covering all training and inference scenarios to maximize resource utilization. • Dedicated compute resources: NPU and other compute resources required for AI development (including development, training, and inference) via the ModelArts console. These are provided through ModelArts resource pools (public and dedicated). • Lite Servers: Cloud-based compute instances provisioned via bare metal servers, accessible via EIPs. • Lite Clusters: Tailored for Kubernetes-based resource users. These provide native Kubernetes APIs, allowing you to directly manage nodes and Kubernetes clusters within the resource pool.
O&M and operations	Integrates O&M monitoring dashboards (Cloud Eye and AOM) to view the detailed status, metrics (such as usage), and logs for training tasks and real-time services.

3 AI Development Basics

3.1 Introduction to the AI Development Lifecycle

What Is AI Development

Artificial intelligence (AI) is a technology capable of simulating human cognition through machines. The core capability of AI is to make a judgment or prediction based on a given input.

What Is the Purpose of AI Development

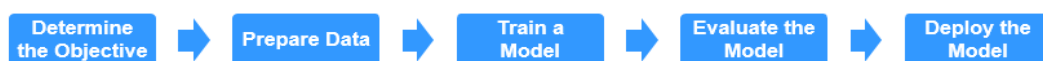
AI development aims to centrally process and extract information from volumes of data to summarize internal patterns of the study objects.

Massive volumes of collected data are computed, analyzed, summarized, and organized by using appropriate statistics, machine learning, and deep learning methods to maximize data value.

Basic Process of AI Development

The basic process of AI development includes the following steps: determining an objective, preparing data, and training, evaluating, and deploying a model.

Figure 3-1 AI development process



Step 1 Determine an objective.

Before starting AI development, determine what to analyze. What problems do you want to solve? What is the business goal? Sort out the AI development framework and ideas based on the business understanding. For example, image classification and object detection. Different projects have different requirements for data and AI development methods.

Step 2 Prepare data.

Data preparation refers to data collection and preprocessing.

Data preparation is the basis of AI development. When you collect and integrate related data based on the determined objective, the most important thing is to ensure the authenticity and reliability of the obtained data. Typically, you cannot collect all the data at the same time. In the data labeling phase, you may find that some data sources are missing and then you may need to repeatedly adjust and optimize the data.

Step 3 Train a model.

Modeling involves analyzing the prepared data to find the causality, internal relationships, and regular patterns, thereby providing references for commercial decision making. After model training, usually one or more machine learning or deep learning models are generated. These models can be applied to new data to obtain predictions and evaluation results.

Developers typically build and train models for their services using popular AI frameworks like TensorFlow, PyTorch, and MindSpore.

Step 4 Evaluate the model.

A model generated by training needs to be evaluated. To achieve a good model, initial results often require refinement through repeated adjustments of algorithm settings and data.

Key metrics like accuracy, recall, and AUC enable effective evaluation and optimization.

Step 5 Deploy the model.

Models are trained using existing data, which can include test data. Once a reliable model is obtained, it is applied to real-world data to make predictions and evaluation, and visualize the results. This information helps decision-makers create effective business strategies by presenting complex insights in an easy-to-understand format.

----End

3.2 Common Concepts of ModelArts

Device-Edge-Cloud

Device-edge-cloud indicates end devices, intelligent edge nodes, and cloud.

Inference

Inference is a process of deriving a new judgment from a known judgment according to a certain strategy. In AI, machines simulate human intelligence and perform inference based on neural networks.

Real-Time Inference

Real-time inference is a web service that synchronously provides inference results for each inference request.

Batch Inference

Batch inference processes batch data for inference.

Ascend Chip

Ascend chips are Huawei-developed AI chips that offer high computing performance while consuming low power.

Resource pool

ModelArts provides large-scale compute clusters for model development, training, and deployment. Both public resource pool and dedicated resource pool are available for you to select.

ModelArts provides a public resource pool by default. ModelArts dedicated resource pools are created separately and used exclusively.

Lite Servers and Lite Clusters are also dedicated.

MoXing

A lightweight distributed framework developed by the ModelArts team and built on deep learning engines such as TensorFlow, PyTorch, and MindSpore. It improves performance of these engines and makes them easier to use. MoXing contains many components. MoXing Framework is a basic common component that can be used to access OBS. It is decoupled from AI engines and can be used in all ModelArts-supported AI engines such as TensorFlow, PyTorch, and MindSpore.

MoXing Framework provides common data file operations in OBS, such as reading, writing, listing, creating folders for, querying, moving, copying, and deleting data files.

You can call MoXing APIs in ModelArts notebook instances without downloading or installing the SDKs. Therefore, MoXing is more convenient than ModelArts and OBS SDKs.

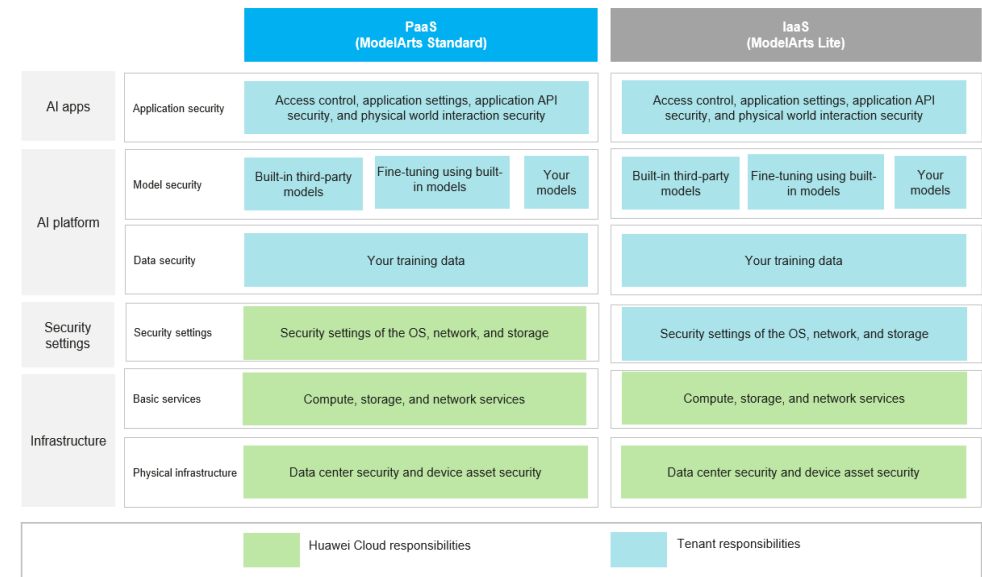
4 Security

4.1 Shared Responsibilities

Huawei guarantees that its commitment to cyber security will never be outweighed by the consideration of commercial interests. To cope with emerging cloud security challenges and pervasive cloud security threats and attacks, Huawei Cloud builds a comprehensive cloud service security assurance system for different regions and industries based on Huawei's unique software and hardware advantages, laws, regulations, industry standards, and security ecosystem.

[Huawei Cloud Security White Paper](#) elaborates on the ideas and measures for building Huawei Cloud security, including cloud security strategies, the shared responsibility model, compliance and privacy, security organizations and personnel, infrastructure security, tenant service and security, engineering security, O&M security, and ecosystem security.

Figure 4-1 Huawei Cloud shared security responsibility model



In the preceding model, different responsibilities are assigned at different layers.

- **AI applications:** You can build diverse AI applications like OCR, NLP, or large language models. You need to secure their application's API connections. For applications linked to physical devices, such as smart machines, consider their real-world safety impacts. You bear full responsibility for ensuring this security.
- **AI platform:** The platform refers to ModelArts. It uses models and training data sourced from open communities or supplied by you. You handle your own security measures.
- **Security configuration:** The security configuration and management depends on the resource mode. For ModelArts dedicated resource pools, **Huawei Cloud** handles resources and ensures security. For Lite resources, you manage resources, configure services, and maintain security.
- **Infrastructure:** **Huawei Cloud** data centers manage the security of essential services like compute, storage, network, and their physical infrastructure across all resource modes.

4.2 Asset Identification and Management

Asset Identification

Your assets in AI Gallery include your published AI assets and your personal information.

AI assets include but are not limited to texts, graphics, data, articles, photos, images, illustrations, code, AI algorithms, and AI models.

Your personal information includes:

- Nickname, profile photo, and email for account registration
- Name, mobile number, and email for participating in practices
- Enterprise information for becoming a partner
- Contact name, mobile number, and email for publishing assets

Asset Management

AI Gallery centrally manages assets published by users.

- AI Gallery stores file assets in official OBS buckets.
- AI Gallery stores image assets in official SWR repositories.

AI Gallery stores personal information of users in databases. AI Gallery encrypts sensitive personal information, such as mobile numbers and emails, in databases.

For more information about AI Gallery, see [AI Gallery](#).

4.3 Identity Authentication and Access Control

Identity Authentication

You can use ModelArts services through the console, APIs, or SDKs. Essentially, access requests are sent through ModelArts REST APIs.

ModelArts APIs can be accessed upon successful authentication. Requests sent through the console can be authenticated using tokens, and requests for calling APIs can be authenticated using tokens or AK/SK. For details, see [Authentication](#).

Access Control

ModelArts allows you to configure fine-grained permissions for refined management of resources and permissions. To do so, ModelArts provides IAM permission control, agency authorization, and workspace.

- IAM permission control
To use ModelArts functions, you need to grant permissions through IAM. For example, if you need to create a training job on ModelArts, you must have the **modelarts:trainJob:create** permission.

If no fine-grained authorization policy is configured for a user created by the administrator, the user has all permissions of ModelArts by default. To control user permissions, the administrator needs to add the user to a user group on IAM and configure fine-grained authorization policies for the user group. In this way, the user obtains the permissions defined in the policies before performing operations on cloud service resources. During policy-based authorization, the administrator can select the authorization scope based on ModelArts resource types. For details about resource permissions, see [Permissions Policies and Supported Actions](#).

- Agency authorization
ModelArts needs to access other services for AI computing. For example, ModelArts needs to access OBS to read your data for training. For security purposes, ModelArts must be authorized to access other cloud services. This is agency authorization.
ModelArts does not save your token authentication credentials. Before performing operations on your resources (such as OBS buckets) in a backend job, you are required to explicitly authorize ModelArts through an IAM agency. ModelArts will use the agency to obtain a temporary authentication credential for performing operations on your resources. For details, see [Configuring Agency Authorization for ModelArts with One Click](#).

- Workspace
Workspace allows customers who have enabled [enterprise projects](#) to divide their resources into multiple logically isolated spaces and control access to different spaces.
After workspace is enabled, a default workspace is created. All resources you have created are in this workspace. A workspace is like a ModelArts twin. You can switch between workspaces in the upper left corner of the navigation pane. Jobs in different workspaces do not affect each other. ModelArts allows you to create multiple workspaces to develop algorithms and manage and deploy models for different service objectives. In this way, the development outputs of different applications are managed in different workspaces for use.

Remote Access Management

When you use a local IDE to remotely access the ModelArts notebook development environment through SSH, the key pair is required for authentication.

4.4 Data Protection

ModelArts takes different measures to keep data stored in ModelArts secure and reliable.

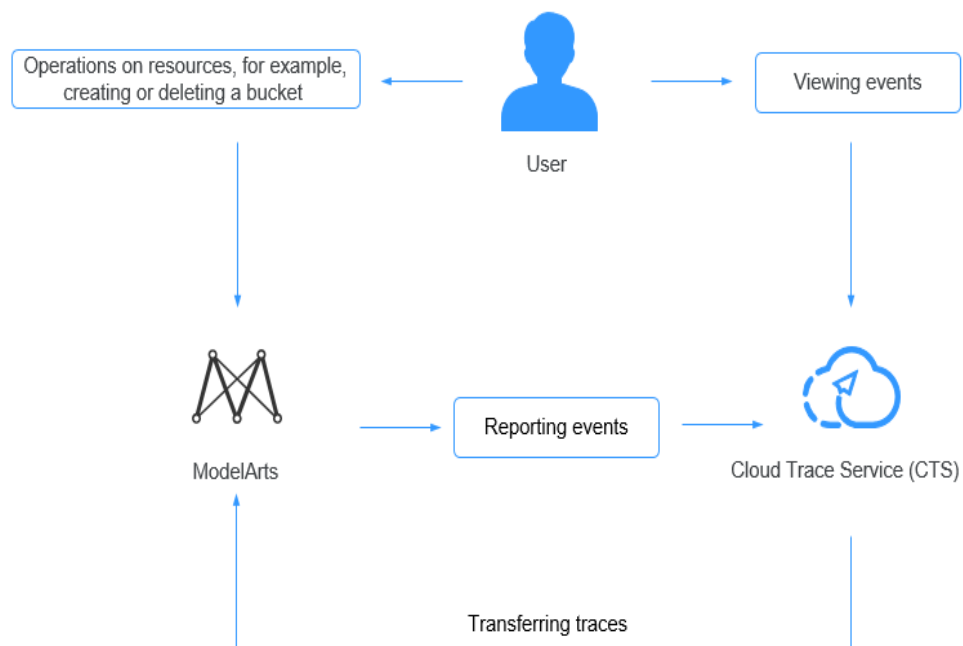
Measure	Description
Static data protection	AI Gallery encrypts sensitive personal information, such as mobile numbers and emails, in databases. The AES encryption algorithm is used.
Protection for data in transit	ModelArts supports importing models via both HTTP and HTTPS protocols, but HTTPS is recommended for more secure data transmission.
Data integrity check	When you upload model files or AI Gallery assets for inference deployment, data may become inconsistent due to network hijacking, caching, and other reasons. ModelArts verifies data consistency by calculating the SHA256 value when data is uploaded or downloaded.
Data isolation mechanism	When a notebook instance is created, data storage of different tenants is isolated, so that different tenants cannot view data of other tenants.

4.5 Auditing and Logs

Auditing

Cloud Trace Service (CTS) records operations on the cloud resources in your account. You can use the logs generated by CTS to perform security analysis, trace resource changes, audit compliance, and locate faults.

After you enable CTS and configure a trace task, CTS can record management and data traces of ModelArts for auditing.

Figure 4-2 CTS

For details about how to enable and configure CTS, see [Enabling CTS](#).

For details about the ModelArts operations traced by CTS, see [ModelArts Standard Key Operations Traced by CTS](#).

Log

You can enable ModelArts logging for analysis or audit. After CTS is enabled, CTS starts recording operations on ModelArts. The CTS management console stores the last seven days of operation records. This section describes how to view operation records of the last seven days on the CTS console.

For details about how to view audit logs on CTS, see [Viewing ModelArts Audit Logs](#).

4.6 Service Resilience

Resilience refers to security resilience of cloud services after attacks, excluding reliability and availability. This chapter describes ModelArts capabilities of defense and detection against intrusions, defense against jitter, proper use of domain names, and content security detection.

Security Suite and Cloud Bastion Host for Enhanced Defense and Detection Against Intrusions

The security suites ModelArts deployed at the host, application, network, and data layers can timely detect security intrusions.

- ModelArts uses web secure components to prevent web security risks from web applications deployed on it and uses WAF for security protection.
- Host Security Service (HSS) products have been deployed on all hosts that carry ModelArts services. These products include but not limited to Huawei-developed HSS and Compute Security Platform (CSP).
- Vulnerability Scan Service (VSS) has been deployed on ModelArts and performs routine scanning to quickly detect and fix vulnerabilities.
- ModelArts performs security O&M on cloud resources through a security management platform.
- Situation Awareness (SA) has been deployed on ModelArts to understand security situation, query attack histories, and promptly detect compliance risks and respond to threat alarms.
- Advanced Anti-DDoS (AAD) has been deployed on the EIPs that carry key ModelArts services to prevent traffic storms.
- Database Security Service (DBSS) has been deployed on ModelArts databases that store important data.

Jitter Prevention and Emergency Response and Restoration Policies Against Attacks

ModelArts isolates resources of different tenants, so that attacks on a tenant's resources will not affect others' resources.

- ModelArts provides dedicated resource pools that are physically isolated, so that attacks on a tenant's resources will not affect others' resources.
- ModelArts defines and maintains its performance specifications to defend attacks, for example, by configuring traffic control on API access.
- ModelArts provides alarm reporting and self-protection against attacks.
- ModelArts detects abnormal service behavior, for example, by detecting abnormal operations platform data and integrating security logs.
- ModelArts provides risk control and emergency response against attacks. For example, ModelArts quickly identifies malicious tenants and malicious IP addresses.
- ModelArts quickly restores services after traffic attacks stop.

Domain Name Usage Specifications and Tenant Content Security Policies of Cloud Services

ModelArts domain names meet certain security requirements to avoid compliance risks and phishing attacks.

Domain names visible to tenants: domain names accessible to tenants, which require more attention to security and compliance.

Domain names invisible to tenants: domain names used by Huawei Cloud services to call each other on the intranet, in which case external users are not able to access the authoritative DNS servers; or domain names that can only be accessed by Huawei employees, partner staff, and outsourced personnel in yellow and green zones through Huawei's office network (namely these domain names cannot be accessed over the Internet).

- Huawei Cloud basic domain names are not directly allocated to tenants but securely used.
- External domain names that have been licensed are not used by Huawei Cloud services to call each other on the intranet.

4.7 Security Risk Monitoring

ModelArts automatically monitors your real-time services and models in real time and manages alarms and notifications, so that you can keep track of performance metrics of services and models. For details, see [Viewing Performance Metrics of a Real-Time Service on Cloud Eye](#).

4.8 Fault Recovery

ModelArts global infrastructure is built for Huawei Cloud regions and AZs. A Huawei Cloud region provides multiple physically independent and isolated AZs that are connected through networks with low latency, high throughput, and high redundancy. You can design and operate faulty applications and databases automatically migrated between AZs without interrupting services. Compared with the traditional infrastructure of a single data center or multiple data centers, AZs provide higher availability, fault tolerance, and scalability.

ModelArts backs up its database data for recovery in case of a service failure or original data damage.

Fault Environment Recovery

If a compute node used by a notebook instance is faulty, the instance will be automatically migrated to another available node. Then, the instance is restored. ModelArts enables you to mount an EVS disk to an instance. Huawei Cloud EVS provides scalable block storage that features high reliability, high performance, and a variety of specifications for servers. Data durability reaches 99.9999999%.

Automatic Recovery from a Training Fault

During model training, a training failure may occur due to a hardware fault. For hardware faults, ModelArts provides fault tolerance check to isolate faulty nodes to improve user experience in training.

The fault tolerance check involves environment pre-check and periodic hardware check. If any fault is detected during either of the checks, ModelArts automatically isolates the faulty hardware and issues the training job again. In distributed training, the fault tolerance check will be performed on all compute nodes used by the training job.

Recovery from an Inference Deployment Fault

During the service running, if an inference instance is faulty due to a hardware fault, ModelArts automatically detects the fault and migrates the faulty instance to another available node. After the instance is restarted, it will be restored. The faulty node is automatically isolated and not be scheduled for running inference instances.

4.9 Update Management

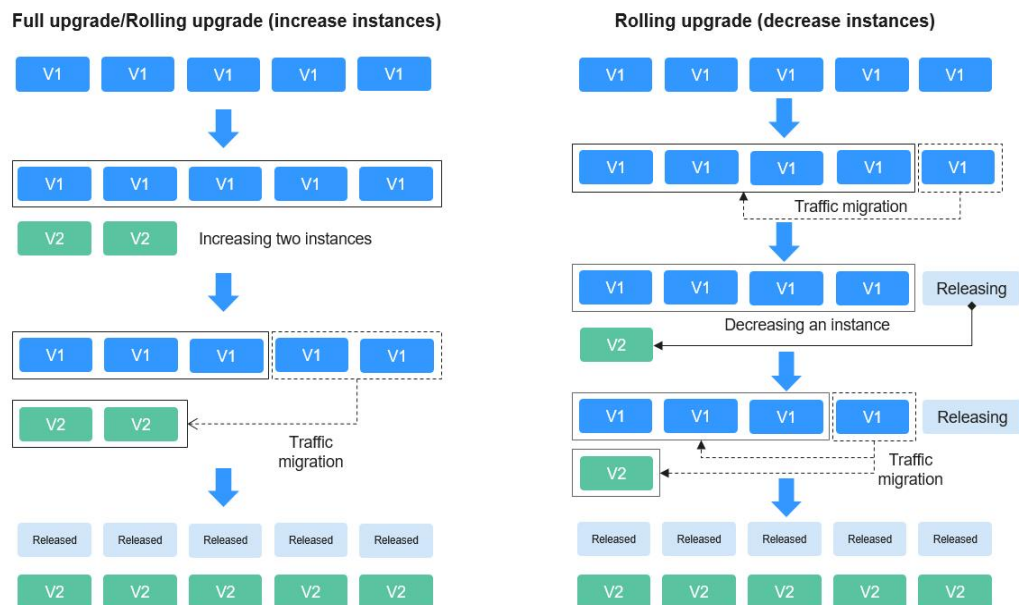
ModelArts Real-Time Service Upgrade

For a deployed service, you can change the model version to upgrade it.

Services can be upgraded in three modes: full upgrade, rolling upgrade (increase instances), and rolling upgrade (decrease instances). For details about the three upgrade modes, see [Figure 4-3](#).

- Full upgrade
Twice the number of resources required by the service will be used to create new-version instances in full mode.
- Rolling upgrade (increase instances)
Extra resources will be used for a rolling upgrade. The more the instances, the faster the upgrade.
- Rolling upgrade (decrease instances)
Certain nodes that were reserved to run services will be used for a rolling upgrade. The more the instances for upgrade, the faster the upgrade, but with a higher probability of service interruption.

Figure 4-3 Service upgrade process



For details about how to upgrade an inference service, see [Modifying a Real-Time Service](#).

Image Upgrade

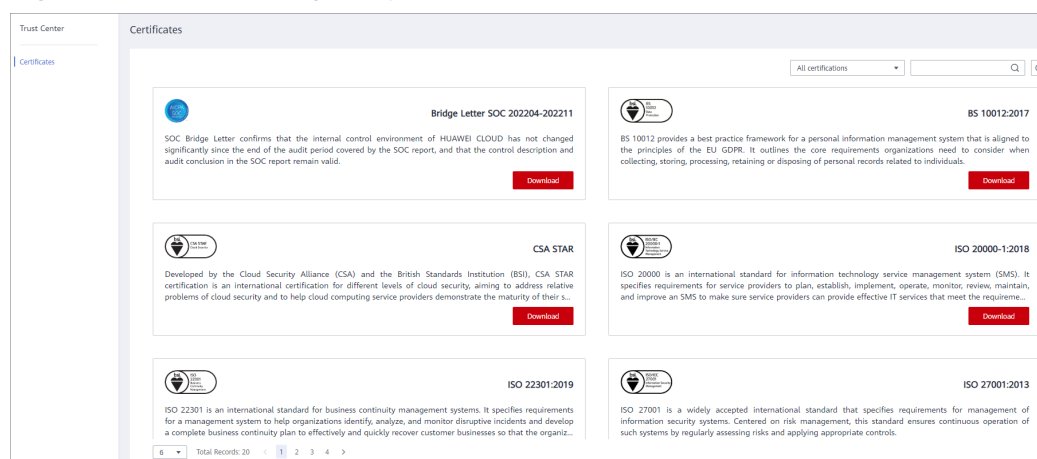
ModelArts provides three function modules: DevEnviron, training management, and inference deployment. The three modules provide base images by the same process. These images are upgraded irregularly to fix vulnerabilities.

4.10 Certificates

Compliance Certificates

Huawei Cloud services and platforms have obtained various security and compliance certifications from authoritative organizations, such as International Organization for Standardization (ISO). You can [download](#) them from the console.

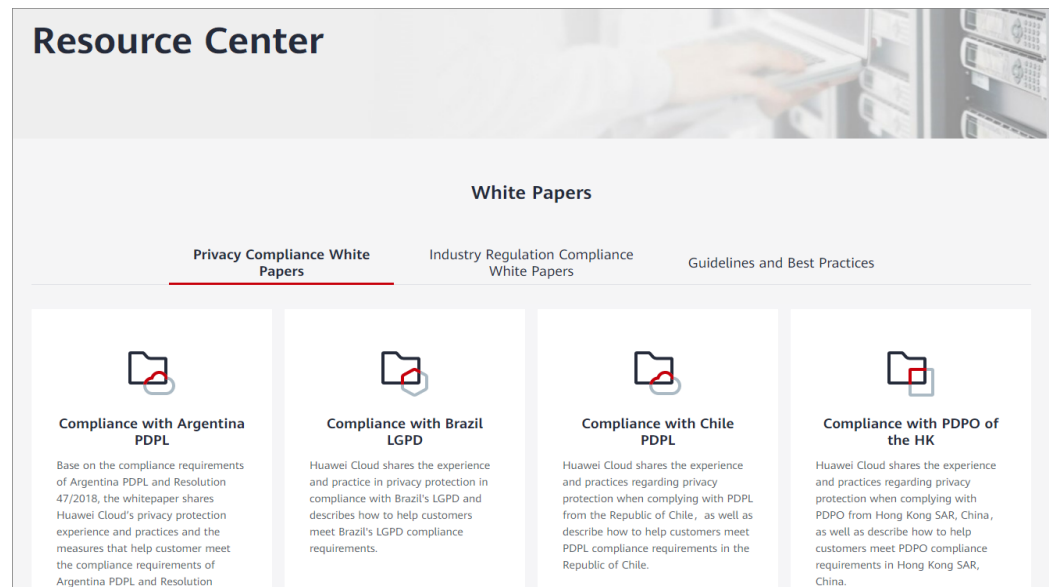
Figure 4-4 Downloading compliance certificates



Resource Center

Huawei Cloud also provides the following resources to help users meet compliance requirements. For details, see [Resource Center](#).

Figure 4-5 Resource center



4.11 Secure Boundaries

The shared responsibility model is a cooperation mode where both providers and customers take security and compliance responsibilities of cloud services.

The providers manage the cloud infrastructure and provide secure hardware and software to ensure the service availability. The customers protect their data and applications, while complying with related compliance requirements.

The providers are responsible for the services and functions and should:

- Establish and maintain secure infrastructure, including networks, servers, and storage devices.
- Provide reliable underlying platforms to ensure runtime security for the environment.
- Provide identity authentication and access control to ensure that only authorized users can access the cloud services and tenants are isolated from each other.
- Provide reliable backup and disaster recovery to prevent data loss due to hardware faults or natural disasters.
- Provide transparent monitoring and incident response services, security updates, and vulnerability patches.

The customers should:

- Encrypt data and applications for confidentiality and integrity.
- Ensure that the model software is securely updated and vulnerabilities are fixed.
- Comply with related regulations, such as GDPR, HIPAA, and PCI DSS.
- Control access to ensure that only authorized users can access and manage resources such as online services.

- Monitor and report any abnormal activity and take actions in a timely manner.

Inference Deployment Security Responsibilities

- Providers
 - Fix the patches related to underlying ECSs.
 - Upgrade the K8S and fix vulnerabilities.
 - Operate VM OS lifecycle maintenance.
 - Ensure the security and compliance of the ModelArts inference platform.
 - Improve the security of containerized application services.
 - Upgrade the model runtime environment and fix vulnerabilities periodically.
- Customers
 - Authorize resource use and control access.
 - Ensure the security of applications, its supply chain, and dependencies by security scanning, auditing, and access verification.
 - Minimize permissions and limit credential delivery.
 - Ensure the security of models (custom images, OBS models, and dependencies) during runtime.
 - Update and fix vulnerabilities in a timely manner.
 - Securely store sensitive data such as credentials.

Best Practices for Inference Deployment Security

- External service authorization

ModelArts inference requires authorization from other cloud services. You can grant only the required permissions based on your needs. For example, you can grant access permission on an OBS bucket to a tenant for model management.
- Internal resource authorization

ModelArts inference supports fine-grained permission control. You can configure the permissions for users based on the actual needs to restrict the permissions on some resources.
- Model management

To decouple models from images and protect model assets, you can dynamically import models from trainings or OBS. You need to upgrade the dependency packages of models, and fix vulnerabilities in open-source or third-party packages. Sensitive information related to models needs to be decoupled and configured during real-time service deployment. Select the runtime environment recommended by ModelArts. The earlier environments may have security vulnerabilities.

You can select open trusted images when creating models using a container image, for example, images from OpenEuler and Ubuntu. Create non-root users rather than root users to run an image. Only the security package required during the runtime is installed in the image. Downsize the image and upgrade the installation package to the latest vulnerability-free version.

Decouple sensitive information from images during service deployment. Ensure that it is not hardcoded in the Dockerfile directly. Perform security scanning on images periodically and install patches to fix vulnerabilities. To facilitate alarm reporting and fault rectification, add health check interface and ensure that the service status can be returned properly. To ensure the service data security, use HTTPS transmission streams and reliable encryption suites for containers.

- **Model deployment**

To prevent services from being overloaded or wasted, set proper compute node specifications during deployment. Do not listen to other ports in the container. If other ports need to be accessed locally, listen to them on localhost. Do not directly transfer sensitive information through environment variables. Encrypt sensitive information with encryption component before data transmission.

App authentication key is an access credential for real-time services. You must keep the app key properly.

ModelArts Security Hardening

ModelArts secures its platform and user services by applying security enhancements to both public pools and dedicated pools. These measures include:

1. SSH access restriction: By default, containers are prevented from accessing port 22.
2. Host NFS service access restriction: By default, containers are blocked from accessing ports 20048 and 2049.
3. Kubernetes API Server access restriction: By default, containers are blocked from accessing port 10250.
4. VXLAN protocol port restriction: By default, containers are blocked from accessing port 4789.
5. OpenStack management address access restriction: By default, containers are blocked from accessing 169.254.169.254.

5 Notes and Constraints

This section describes some limitations and constraints on using ModelArts.

Specifications Restrictions

Table 5-1 Specifications description

Resource Type	Specifications	Description
Compute resources	All compute resource specifications in pay-per-use, yearly/monthly, and package modes, including CPU, GPU, and NPU	All types of purchased compute resources cannot be used across regions.
Compute resources	Package	Packages are used only for public resource pools and cannot be used for dedicated resource pools.

Constraints

Table 5-2 Function constraints

Item	Constraints
Dedicated resource pool	- It is good practice to create no more than 30 nodes at a time. Otherwise, the creation may fail due to traffic limiting. For more information, see Creating a Standard Dedicated Resource Pool. - Only pools in the Running state can be resized. The number of instances cannot be decreased to 0. - A pool's job types can only be modified while it is running. - A pool's driver can only be upgraded while it is running and there are GPU or Ascend resources in its nodes. - For a logical resource pool, the driver can be upgraded only after node binding is enabled. To enable node binding, submit a service ticket to contact engineers.
Notebook	- Deleted notebook instances cannot be recovered. After a notebook instance is deleted, the data stored in the mounted directory will be deleted. - You can only change an image on a stopped notebook instance. - You can modify a notebook instance's specifications while it is stopped, running, or failed to start. - The target notebook instance must use EVS for storage. If the original capacity of an EVS disk is 4096 GB, the disk capacity cannot be expanded. A maximum of 100 GB can be added at a time. - After the notebook instance is stopped, the expanded EVS capacity still takes effect. The EVS billing is based on the expanded capacity. An EVS disk is billed as long as it is used. To stop billing an EVS disk, delete data from the EVS disk and release the disk. - Images stored in a notebook instance cannot be larger than 35 GB and there cannot be more than 125 image layers. Otherwise, the image cannot be saved.

Item	Constraints
Training job	- Training logs are retained for only 30 days. To permanently store logs, enable persistent log saving and set a job log path for dumping when creating a training job. For Ascend training, you need to configure the OBS path for storing training logs by default. You need to manually enable Persistent Log Saving for training jobs using other resources. - Only dedicated resource pools allow logging in to training containers using Cloud Shell. The training job must be running. - Algorithms subscribed from AI Gallery cannot be saved as new algorithms. - Suspension can be detected only for training jobs that run on GPUs. - The priority can be set for a training job only when a new-version dedicated resource pool is used. The job priority cannot be set for training jobs using a public resource pool or old-version dedicated resource pool. - Only the PyTorch and MindSpore frameworks can be used for distributed training and debugging. If you want to use MindSpore, each node must be equipped with eight cards. - When using a custom image to create a training job, ensure that the custom image size is under 15 GB and does not exceed half of the container engine space in the resource pool. An oversized image affects the startup of a training job. The container engine space of a ModelArts public resource pool is 50 GB. By default, the container engine space of a dedicated resource pool is also 50 GB. You can customize the container engine space when creating a dedicated resource pool. - The uid of the default user of a custom training image must be 1000.
Inference model	- The maximum size of a model file from OBS is 20 GB. For more information, see Creating an AI Application. - If the size of your file exceeds the container engine space, a message will be displayed, indicating that the image space is insufficient. The maximum container engine space in a public resource pool is 50 GB, and that for a dedicated resource pool is 50 GB by default. You can set the container engine space for a dedicated resource pool when you create it, which does not increase costs. For more information, see Restrictions on the Size of an Image for Importing an AI Application. - After deploying a model in an ExeML project, it is automatically added to the model list. ExeML-generated models can only be deployed, not downloaded.

Item	Constraints
Inference service deployment	- Cloud Shell can only access a container when the associated real-time service is deployed within a dedicated resource pool and running. - Batch services can only be deployed in public resource pools. - For models in synchronous request mode, if the prediction request latency exceeds 60 seconds, the request will fail, and there is a possibility that the service may be interrupted. In this case, create an image in asynchronous request mode.
Lite Server	- If a ModelArts Lite Server runs on BMSs, upgrading or changing the OS kernel or driver can render the driver and kernel incompatible, preventing the OS from starting or making basic functions unavailable. To upgrade or change the OS kernel or driver, contact Huawei Cloud technical support. - ModelArts Lite Servers running on ECSs does not support OS reinstallation. Similarly, some BMSs in specific regions have this limitation. To reinstall the OS, switch to a different one. - When you reinstall or change the OS of a ModelArts Lite Server, the EVS system disk ID changes, which may differ from the original ID in your purchase order. This prevents you from expanding the EVS system disk capacity, resulting in an error message: "The order is expired. The capacity cannot be expanded. Renew the order." You are advised to expand the storage capacity by attaching EVS or SFS disks.
Lite Cluster	- Only pools in the Running state can be resized. The number of instances cannot be decreased to 0. - You can specify the container engine space size when creating a resource pool. - To change the container engine space for new nodes in an existing Lite Cluster resource pool, set the desired size. The container engine space of existing nodes cannot be modified, as this would create inconsistencies in the dockerBaseSize across nodes with the same flavor within the pool, potentially disrupting task execution on different nodes. - A pool's driver can only be upgraded while it is running and there are GPU or Ascend resources in its nodes. - For a logical resource pool, the driver can be upgraded only after node binding is enabled. To enable node binding, submit a service ticket to contact engineers.

Item	Constraints
Interaction between ModelArts and OBS	- ModelArts does not support encrypted OBS buckets. When creating an OBS bucket, do not enable bucket encryption. - ModelArts does not support cross-region access to OBS buckets. Ensure that OBS and ModelArts are in the same region.

6 Permissions Management

ModelArts allows you to configure fine-grained permissions for refined management of resources and permissions. This is commonly used by large enterprises, but it is complex for individual users. It is recommended that individual users configure permissions for using ModelArts by referring to [Configuring Agency Authorization](#).

NOTE

Do I need to read this document?

Read this document if any of the following descriptions matches your situation.

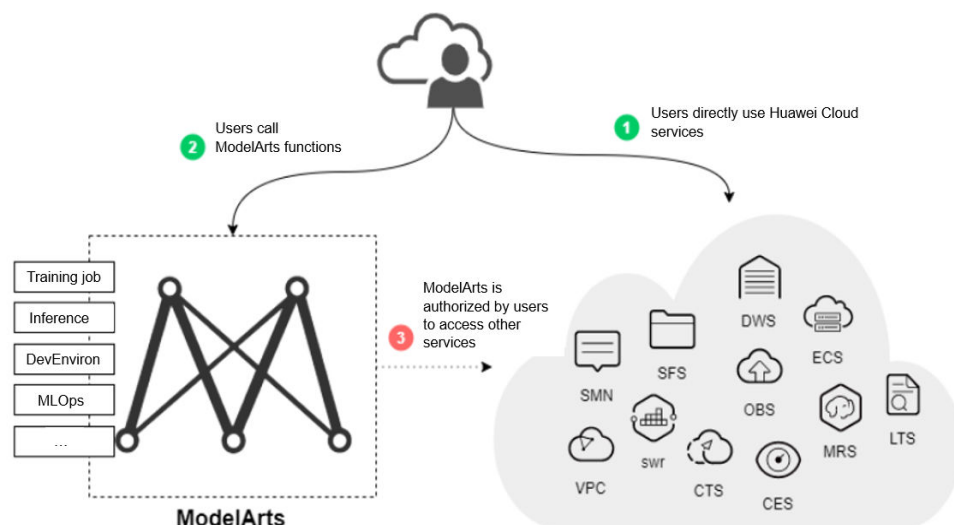
- You are an enterprise user, and
 - There are multiple departments in your enterprise, and you need to control users' permissions so that users in different departments can access only their dedicated resources and functions.
 - There are multiple roles (such as administrators, algorithm developers, and application O&M personnel) in your enterprise. You need them to use only specific functions.
 - There are logically multiple environments (such as the development environment, pre-production environment, and production environment) and are isolated from each other. You need to control users' permissions on different environments.
 - You need to control permissions of specific IAM user or user group.
- You are an individual user, and you have created multiple IAM users. You need to assign different ModelArts permissions to different IAM users.
- You need to understand the concepts and operations of ModelArts permissions management.

ModelArts uses Identity and Access Management (IAM) for most permissions management functions. Before reading below, learn about [Basic Concepts](#). This helps you better understand this document.

To implement fine-grained permissions management, ModelArts provides permission control, agency authorization, and workspace. The following describes the details.

ModelArts Permissions and Agencies

Figure 6-1 Permissions management



ModelArts functions are controlled through IAM permissions. For example, if you, as an IAM user, need to create a training job on ModelArts, you must have the **modelarts:trainJob:create** permission. For details about how to assign permissions to a user (you need to add the user to a user group and then assign permissions to the user group), see [Permissions Management](#).

ModelArts must access other services for AI computing. For example, ModelArts must access OBS to read your data for training. For security purposes, ModelArts must be authorized to access other cloud services. This is agency authorization.

The following summarizes permissions management:

- Your access to any cloud service is controlled through IAM. You must have the permissions of the cloud service. (The required service permissions vary depending on the functions you use.)
- To use ModelArts functions, you need to grant permissions through IAM.
- ModelArts must be authorized by you to access other cloud services for AI computing.

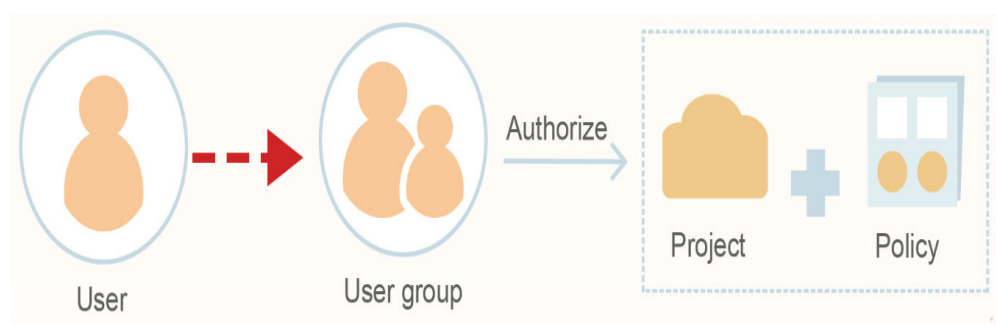
ModelArts Permissions Management

By default, new IAM users do not have any permissions assigned. You need to add the users to a user group and grant the user group with policies, so that the users in the group can inherit the permissions. After authorization, users can perform operations on ModelArts based on permissions.

CAUTION

ModelArts is a project-level service deployed and accessed in specific physical regions. When you authorize an agency, you can set the scope for the permissions you select to all resources, enterprises projects, or region-specific projects. If you specify region-specific projects, the selected permissions will be applied to resources in these projects.

ModelArts supports enterprise projects. You can specify an enterprise project when selecting the authorization scope. For details, see [Creating a User Group and Assigning Permissions](#).



When assigning permissions to a user group, IAM does not directly assign specific permissions to the user group. Instead, IAM adds the permissions to a policy and then assigns the policy to the user group. To facilitate user permissions management, each cloud service provides some preset policies for you to directly use. If the preset policies cannot meet your requirements of fine-grained permissions management, you can customize policies.

Table 6-1 lists all the preset system-defined policies supported by ModelArts.

Table 6-1 System-defined policies supported by ModelArts

Policy	Description	Type
ModelArts FullAccess	Administrator permissions for ModelArts. Users granted these permissions can operate and use ModelArts.	System-defined policy
ModelArts CommonOperations	Common user permissions for ModelArts. Users granted these permissions can operate and use ModelArts, but cannot manage dedicated resource pools.	System-defined policy
ModelArts Dependency Access	Permissions for common dependent services of ModelArts	System-defined policy

Generally, ModelArts FullAccess is assigned only to administrators. If fine-grained management is not required, assigning ModelArts CommonOperations to all users

will meet the development requirements of most small teams. If you want to customize policies for fine-grained permissions management, see [IAM](#).

NOTE

When you assign ModelArts permissions to a user, the system does not automatically assign the permissions of other services to the user. This ensures security and prevents unexpected unauthorized operations. In this case, however, you must separately assign permissions of different services to users so that they can perform some ModelArts operations.

For example, if an IAM user needs to use OBS data for training and the ModelArts training permission has been configured for the IAM user, the IAM user still needs to be assigned with the OBS read, write, and list permissions. The OBS list permission allows you to select the training data path on ModelArts. The read permission is used to preview data and read data for training. The write permission is used to save training results and logs.

- For individual users or small organizations, it is a good practice to configure the **Tenant Administrator** policy that applies to global services for IAM users. In this way, IAM users can obtain all user permissions except IAM. However, this may cause security issues. (For an individual user, its default IAM user belongs to the **admin** user group and has the **Tenant Administrator** permission.)
- If you want to restrict user operations, configure the minimum permissions of OBS for ModelArts users. For details, see [OBS Permissions Management](#). For details about fine-grained permissions management of other cloud services, see the corresponding cloud service documents.

ModelArts Agency Authorization

ModelArts must be authorized by users to access other cloud services for AI computing. In the IAM permission system, such authorization is performed through agencies.

For details about the basic concepts and operations of agencies, see [Cloud Service Delegation](#).

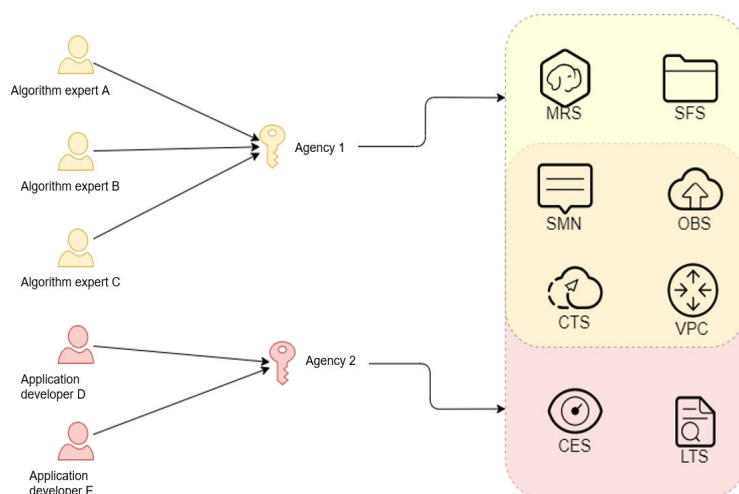
To simplify agency authorization, ModelArts supports automatic agency authorization configuration. You only need to configure an agency for yourself or specified users on the **Global Configuration** page of the ModelArts console.

NOTE

- Only users with the IAM agency management permission can perform this operation. Generally, members in the IAM admin user group have this permission.
- ModelArts agency authorization is region-specific, which means that you must perform agency authorization in each region you use.

On the **Global Configuration** page of the ModelArts console, after you click **Add Authorization**, you can configure an agency for a specific user or all users. Generally, an agency named **modelarts_agency_<Username>_Random ID** is created by default. In the **Permissions** area, you can select the preset permission configuration or select the required policies. If both options cannot meet your requirements, you can create an agency on the IAM management page (you need to delegate ModelArts to access your resources), and then use an existing agency instead of adding an agency on the **Add Authorization** page.

ModelArts associates multiple users with one agency. This means that if two users need to configure the same agency, you do not need to create an agency for each user. Instead, you only need to configure the same agency for the two users.

Figure 6-2 Mapping between users and agencies**NOTE**

A user can use ModelArts only after being associated with an agency. However, even if the permissions assigned to the agency are insufficient, no error is reported when the API is called. An error occurs only when the system uses unauthorized functions. For example, you enable message notification when creating a training job. Message notification requires SMN authorization. However, an error occurs only when messages need to be sent for the training job. The system ignores some errors, and other errors may cause job failures. When you implement permission minimization, ensure that you will still have sufficient permissions for the required operations on ModelArts.

Strict Authorization

In strict authorization mode, explicit authorization by the account administrator is required for IAM users to access ModelArts. The administrator can add the required ModelArts permissions to common users through authorization policies.

In non-strict authorization mode, IAM users can use ModelArts without explicit authorization. The administrator needs to configure the deny policy for IAM users to prevent them from using some ModelArts functions.

The administrator can change the authorization mode on the **Global Configuration** page.

NOTICE

The strict authorization mode is recommended. In this mode, IAM users must be authorized to use ModelArts functions. In this way, the permission scope of IAM users can be accurately controlled, minimizing permissions granted to IAM users.

Managing Resource Access Using Workspaces

Workspace enables enterprise customers to split their resources into multiple spaces that are logically isolated and to manage access to different spaces. As an enterprise user, you can submit the request for enabling the workspace function to your technical support manager.

After workspace is enabled, a default workspace is created. All resources you have created are in this workspace. A workspace is like a ModelArts twin. You can switch between workspaces in the upper left corner of the navigation pane. Jobs in different workspaces do not affect each other.

When creating a workspace, you must bind it to an enterprise project. Multiple workspaces can be bound to the same enterprise project, but one workspace cannot be bound to multiple enterprise projects. You can use workspaces for refined restrictions on resource access and permissions of different users. The restrictions are as follows:

- Users must be authorized to access specific workspaces (this must be configured on the pages for creating and managing workspaces). This means that access to AI assets such as datasets and algorithms can be managed using workspaces.
- In the preceding permission authorization operations, if you set the scope to enterprise projects, the authorization takes effect only for workspaces bound to the selected projects.

 NOTE

- Restrictions on workspaces and permission authorization take effect at the same time. That is, a user must have both the permission to access the workspace and the permission to create training jobs (the permission applies to this workspace) so that the user can submit training jobs in this workspace.
- If you have enabled an enterprise project but have not enabled a workspace, all operations are performed in the default enterprise project. Ensure that the permissions on the required operations apply to the default enterprise project.
- The preceding restrictions do not apply to users who have not enabled any enterprise project.

Summary

Key features of ModelArts permissions management:

- If you are an individual user, you do not need to consider fine-grained permissions management. Your account has all permissions to use ModelArts by default.
- All functions of ModelArts are controlled by IAM. You can use IAM authorization to implement fine-grained permissions management for specific users.
- All users (including individual users) can use specific functions only after agency authorization on ModelArts (**Settings > Add Authorization**). Otherwise, unexpected errors may occur.
- If you have enabled the enterprise project function, you can also enable ModelArts workspace and use both basic authorization and workspace for refined permissions management.

7 Billing Description

ModelArts is a one-stop AI development platform geared toward developers and data scientists of all skill levels. It enables you to rapidly build, train, and deploy models anywhere (from the cloud to the edge), and manage full-lifecycle AI workflows. ModelArts accelerates AI development and fosters AI innovation with key capabilities, including data preprocessing and auto labeling, distributed training, automated model building, and one-click workflow executing.

ModelArts can be billed either pay-as-you-go or on a more economical yearly/monthly basis. For more details, see [Product Pricing Details](#).

For more information, see [ModelArts Billing Modes](#).

8 Quotas

This section describes the quota limits of cloud services related to ModelArts, helping you view and manage your quotas.

What Are Quotas?

A quota defines the maximum number of resources of a certain type that can be created in a region.

To help you save quotas, Huawei Cloud sets limit on the maximum number of cloud resources that you can create in each region.

If the existing resource quotas cannot meet your needs, you can request higher quotas.

Viewing Quotas

Log in to the console to view default quotas. For details, see [Quotas](#).

Increasing Quotas

To increase the resource quota, see [How Do I Apply for a Higher Quota?](#)

Quota Items

To use a ModelArts Lite Cluster or Lite Server, you will need more resources than Huawei Cloud's default quotas provided. This includes more ECS instances, memory, CPU cores, and EVS disk space. You will need to request a higher quota to meet these needs. The following table lists quota items.

Table 8-1 Resource quotas involved in ModelArts Lite Clusters or Lite Servers

Service	Resource Type
ECS resource type	ECS instances
	CPU cores
	RAM capacity (MB)

Service	Resource Type
EIP resources	Bandwidth scaling policies
EVS and SFS resources	Disks
	Disk capacity (GB)
	Snapshots
SFS resources	Capacity limit

When you use a ModelArts dedicated resource pool, the following quotas are involved. If the quotas are insufficient, increase quotas.

Table 8-2 Quotas involved in ModelArts dedicated resource pools

Service	Resource Type	Default Quota
ModelArts	Resource pool	15
	Network	15
	Resource pool instances	2,048
	Resource pool cores	20,480
	Resource pool memory capacity	24,576 GiB
	Resource pool disks	400
	Resource pool disk capacity	32,768 GiB
	snt9b	0

Table 8-3 Quota limits of each function module

Resource Type	Default Quota	Adjustable	Description
Notebook	A maximum of 10 notebook instances can be created under one account.	No	For more information, see Creating a Notebook Instance .
Real-time service	A maximum of 20 real-time services can be created under one account.	Yes Submit a service ticket to increase the quota.	For more information, see Deploying a Model as a Real-Time Service .

Resource Type	Default Quota	Adjustable	Description
Batch service	A maximum of 1,000 batch services can be created under one account.	No	For more information, see Deploying a Model as a Batch Inference Service .
Edge service	A maximum of 1,000 edge services can be created under one account.	No	None
Dedicated resource pool	A maximum of 50 dedicated resource pools can be created under one account.	Yes Submit a service ticket to increase the quota.	For more information, see Creating a Standard Dedicated Resource Pool .
Tag	A maximum of 20 tags can be added to a training job, notebook instance, or real-time service.	No	For more information, see How Does ModelArts Use Tags to Manage Resources by Group?

9 ModelArts and Other Services

IAM

ModelArts uses Identity and Access Management (IAM) for authentication and authorization. For more information about IAM, see the [Identity and Access Management Documentation](#).

OBS

ModelArts does not provide data storage. Instead, it uses Object Storage Service (OBS) to store data, and backs up and takes snapshots for models, achieving secure, reliable, and cost-effective storage. For more information about OBS, see [Object Storage Service Documentation](#).

All the input data, output data, and cache data during AI development can be stored in OBS buckets for reading.

Figure 9-1 Interaction between ModelArts and OBS

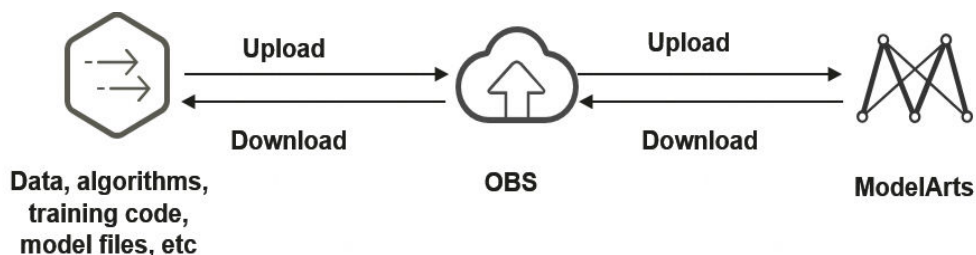


Table 9-1 Relationship between ModelArts and OBS

Function	Task	Relationship
AI development lifecycle	Data management	• Datasets are stored in OBS. • Dataset labeling information is stored in OBS. • Data can be imported from OBS.
	Development environment	Data or code files in a notebook instance are stored in OBS.

Function	Task	Relationship
	Model training	• The datasets used by training jobs are stored in OBS. • The running scripts for training jobs are stored in OBS. • The models generated by training jobs are stored in the specified OBS paths. • The run logs of training jobs are stored in the specified OBS paths.
	Model management	After a training job is completed, the generated model is stored in OBS. You can import the model from OBS.
	Model deployment	Models stored in OBS can be deployed as services.

EVS

ModelArts uses Elastic Volume Service (EVS) to store created notebook instances. For details, see [Elastic Volume Service User Guide](#).

CCE

ModelArts deploys models as real-time services using Cloud Container Engine (CCE), which supports high concurrency and auto scaling. For more information about CCE, see [Cloud Container Engine User Guide](#).

SWR

To use an AI framework that is not supported by ModelArts, use Software Repository for Container (SWR) to customize an image and import the image to ModelArts for training or inference. For details about SWR, see [Software Repository for Container User Guide](#).

Cloud Eye

ModelArts uses Cloud Eye to monitor real-time services and model loads in real time and send alarms and notifications automatically. For details about Cloud Eye, see [Cloud Eye User Guide](#).